

Power BI Assignment 1 – Data Transformation & Data Modelling

Here, the screenshots of data transformation and data modelling steps,

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Merge

Select tables and matching columns to create a merged table.

List of Orders

Order ID	Order Date	CustomerName	State	City	Location
B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
B-25602	4/1/2018	Pearl	Maharashtra	Pune	Pune, Maharashtra
B-25603	4/3/2018	Jahan	Madhya Pradesh	Bhopal	Bhopal, Madhya Pradesh
B-25604	4/3/2018	Divsha	Rajasthan	Jaipur	Jaipur, Rajasthan
B-25605	4/5/2018	Kasheen	West Bengal	Kolkata	Kolkata, West Bengal

Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
B-25601	1,275.00	-1148	7	Furniture	Bookcases	-90.04 %	Loss
B-25601	66.00	-12	5	Clothing	Stole	-18.18 %	Loss
B-25601	8.00	-2	3	Clothing	Hankerchief	-25.00 %	Loss
B-25601	80.00	-56	4	Electronics	Electronic Games	-70.00 %	Loss
B-25602	168.00	-111	2	Electronics	Phones	-66.07 %	Loss

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

☐ Fuzzy matching options

✓ The selection matches 500 of 500 rows from the first table.

OK Cancel

6 COLUMNS, 500 ROWS Column profiling based on top 1000 rows

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Table.RenameColumns

Table.RenameColumns(removedColumns, newColumnNames)

removedColumns: {"Order Details.Amount", "Amount"}, {"Order Details.Profit", "Profit"}

newColumnNames: {"Order Details.Amount", "Amount"}, {"Order Details.Profit", "Profit"}

Orders Data

Order ID	Order Date	CustomerName	State	City	Location
1 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
2 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
3 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
4 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
5 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
6 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
7 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
8 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
9 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
10 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
11 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
12 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
13 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
14 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
15 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
16 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
17 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
18 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
19 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
20 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
21 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
22 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
23 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat
24 B-25601	4/1/2018	Bharat	Gujarat	Ahmedabad	Ahmedabad, Gujarat

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Power BI Assignment 1 – Data Transformation & Data Modelling

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File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Date Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Text Analytics Vision Azure Machine Learning

Queries [4] List of Orders Order Details Sales target Orders Data

fx = Table.SelectRows(*Renamed Columns", each true)

Order ID Order Date CustomerName State City Location

Sort Ascending Sort Descending Clear Sort Clear Filter Remove Empty Date Filters

Search

(Select All) 4/1/2018 4/3/2018 4/5/2018 4/6/2018 4/8/2018 4/9/2018 4/11/2018 4/12/2018 4/13/2018 4/15/2018 4/17/2018 4/18/2018 4/20/2018

OK Cancel

13 COLUMNS, 999+ ROWS Column

Query Settings

PROPERTIES Name Orders Data All Properties

APPLIED STEPS Source Expanded Order Details Removed Columns Renamed Columns Filtered Rows

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Date Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Text Analytics Vision Azure Machine Learning

Queries [4] List of Orders Order Details Sales target Orders Data

fx = Table.Sort(*Filtered Rows",{{"Order Date", Order.Descending}})

Order ID Order Date CustomerName State City Location

1 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

2 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

3 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

4 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

5 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

6 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

7 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

8 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

9 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

10 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

11 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

12 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

13 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

14 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

15 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

16 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

17 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

18 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

19 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

20 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

21 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

22 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

23 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

24 B-26100 3/31/2019 Hitika Madhya Pradesh Indore Indore, Madi

13 COLUMNS, 999+ ROWS Column

Column sorting based on top 1000 rows

Query Settings

PROPERTIES Name Orders Data All Properties

APPLIED STEPS Source Expanded Order Details Removed Columns Renamed Columns Filtered Rows Sort Descending In Order Date Sort Descending In Order Date

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Power BI Assignment 1 – Data Transformation & Data Modelling

The screenshot shows the Power Query Editor interface. The main data table has columns: Order ID, Order Date, CustomerName, State, City, and Location. A context menu is open over the 'Order Date' column, with the 'Sort Descending' option selected. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Sort Descending In Order Date'. The status bar at the bottom indicates '13 COLUMNS, 999+ ROWS'.

Order ID	Order Date	CustomerName	State	City	Location
1	3/31/2				
2	3/31/2				
3	3/31/2				
4	3/31/2				
5	3/31/2				
6	3/31/2				
7	3/31/2				
8	3/31/2				
9	3/31/2				
10	3/31/2				
11	3/31/2				
12	3/31/2				
13	3/31/2				
14	3/31/2				
15	3/31/2				
16	3/31/2				
17	3/31/2				
18	3/31/2				
19	3/31/2				
20	3/31/2				
21	3/31/2				
22	3/31/2				
23	3/31/2				
24	3/31/2				

The screenshot shows the Power Query Editor interface after filtering. The main data table now only contains rows where the 'State' is 'Tamil Nadu'. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list, which includes 'Filtering Rows in State Column as Tamilnadu'. The status bar at the bottom indicates '13 COLUMNS, 245 ROWS'. A tooltip 'Filtering Rows in State Column as Tamilnadu' is visible over the 'State' column header.

Order ID	Order Date	CustomerName	State	City	Location
1	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
2	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
3	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
4	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
5	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
6	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
7	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
8	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
9	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
10	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
11	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
12	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
13	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
14	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
15	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
16	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
17	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
18	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
19	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
20	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
21	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
22	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
23	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan
24	3/22/2019	Aarushi	Tamil Nadu	Chennai	Chennai, Tan

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The screenshot shows the Power Query Editor interface. The main area displays a table with two columns: 'Order ID' and 'Order ID Count'. The 'Order ID' column contains values from B-25601 to B-25624. The 'Order ID Count' column contains numerical values ranging from 1 to 6. The formula bar at the top shows the query definition: `= Table.Group(#"Changed Type1", {"Order ID"}, {"Order ID Count", each Table.RowCount(_), Int64.Type})`. The right-hand pane shows the 'Query Settings' for 'Order Details (2)', with the 'APPLIED STEPS' list including 'Count of Order ID'. The status bar at the bottom indicates '2 COLUMNS, 500 ROWS' and 'Column profiling based on top 1000 rows'.

Order ID	Order ID Count
B-25601	4
B-25602	5
B-25603	8
B-25604	2
B-25605	1
B-25606	1
B-25607	1
B-25608	4
B-25609	2
B-25610	6
B-25611	1
B-25612	1
B-25613	1
B-25614	2
B-25615	1
B-25616	4
B-25617	1
B-25618	2
B-25619	1
B-25620	1
B-25621	3
B-25622	1
B-25623	4
B-25624	1

The screenshot shows the Power Query Editor interface. The main area displays a table with two columns: 'Category' and 'Average Profit'. The 'Category' column contains values 'Furniture', 'Clothing', and 'Electronics'. The 'Average Profit' column contains numerical values: 9.456790123, 11.76290832, and 34.07142857. The formula bar at the top shows the query definition: `= Table.Group(#"Changed Type in Profit Status", {"Category"}, {"Average Profit", each List.Average([Profit])`. The right-hand pane shows the 'Query Settings' for 'Order Details (3)', with the 'APPLIED STEPS' list including 'Grouped Rows in average pro...'. The status bar at the bottom indicates '2 COLUMNS, 3 ROWS' and 'Column profiling based on top 1000 rows'.

Category	Average Profit
Furniture	9.456790123
Clothing	11.76290832
Electronics	34.07142857

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Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Data Sources Parameters Refresh Preview Advanced Editor Manage Columns Remove Columns Keep Rows Remove Rows Split Column Group By Use First Row as Headers Replace Values Data Type: Decimal Number Merge Queries Append Queries Combine Files Text Analytics Vision Azure Machine Learning

Queries [7]

Table.Group(#"Changed Type in Profit Status", {"Sub-Category"}, {"Total Amount", each List.Sum([Amount]), type text})

Sub-Category	Total Amount
Bookcases	56861
Stole	18546
Hankerchief	14608
Electronic Games	39168
Phones	46119
Saree	53511
Trousers	30039
Chairs	34222
Kurti	3361
T-shirt	7382
Shirt	7555
Leggings	2106
Tables	22614
Printers	58252
Accessories	21728
Furnishings	13484
Skirt	1946

2 COLUMNS, 17 ROWS Column profiling based on top 1000 rows

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Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Column From Custom Examples Column Invoke Custom Function Duplicate Column Conditional Column Index Column Merge Columns Extract Trigonometry Rounding Information Date Time Duration Text Analytics Vision Azure Machine Learning

Queries [8]

Table.AddColumn(#"Inserted Month", "Month Name", each Date.MonthName([Month of Order Date]), type text)

Month of Order Date	Category	Target	Month	Month Name
4/18/2026	Furniture	10,400.00	4	April
5/18/2026	Furniture	10,500.00	5	May
6/18/2026	Furniture	10,600.00	6	June
7/18/2026	Furniture	10,800.00	7	July
8/18/2026	Furniture	10,900.00	8	August
9/18/2026	Furniture	11,000.00	9	September
10/18/2026	Furniture	11,100.00	10	October
11/18/2026	Furniture	11,300.00	11	November
12/18/2026	Furniture	11,400.00	12	December
1/19/2026	Furniture	11,500.00	1	January
2/19/2026	Furniture	11,600.00	2	February
3/19/2026	Furniture	11,800.00	3	March
4/18/2026	Clothing	12,000.00	4	April
5/18/2026	Clothing	12,000.00	5	May
6/18/2026	Clothing	12,000.00	6	June
7/18/2026	Clothing	14,000.00	7	July
8/18/2026	Clothing	14,000.00	8	August
9/18/2026	Clothing	14,000.00	9	September
10/18/2026	Clothing	16,000.00	10	October
11/18/2026	Clothing	16,000.00	11	November
12/18/2026	Clothing	16,000.00	12	December
1/19/2026	Clothing	16,000.00	1	January
2/19/2026	Clothing	16,000.00	2	February
3/19/2026	Clothing	16,000.00	3	March

5 COLUMNS, 36 ROWS Column profiling based on top 1000 rows

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Power BI Assignment 1 – Data Transformation & Data Modelling

The screenshot shows the Power Query Editor window titled "Untitled - Power Query Editor". The ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The main area displays a table with the following data:

Month of Order Date	1.2 Total Target Amount
4/18/2026	31400
5/18/2026	31500
6/18/2026	31600
7/18/2026	33800
8/18/2026	33900
9/18/2026	34000
10/18/2026	36100
11/18/2026	36300
12/18/2026	36400
1/19/2026	43500
2/19/2026	43600
3/19/2026	43800

The right-hand pane shows the "Query Settings" for "Sales target (2)", including the "PROPERTIES" and "APPLIED STEPS" sections. The "APPLIED STEPS" list includes: Source, Promoted Headers, Changed Type, Inserted Month, Inserted Month Name, and Grouped Rows. The bottom status bar indicates "2 COLUMNS, 12 ROWS" and "Column profiling based on top 1000 rows".

The screenshot shows the Power BI Desktop window titled "Untitled - Power BI Desktop". The ribbon includes tabs for File, Home, and Help. The main area displays a data model with two tables: "List of Orders" and "Order Details".

List of Orders Table:

- CustomerName
- Location
- Order Date
- Order ID

Order Details Table:

- Amount
- Category
- Order ID
- Profit
- Profit Margin
- Profit Status

The tables are connected by a relationship line. The "List of Orders" table is connected to the "Order Details" table via the "Order ID" field. The relationship is marked with a "1" on the "List of Orders" side and a "*" on the "Order Details" side, indicating a one-to-many relationship. The bottom status bar shows "All tables" and "Update available (click to download)".

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